



Rule-Based Enterprise Workflow Optimization for Identifying and Reducing Task Bottlenecks Using Functions and Dynamic Memory Allocation

CSA0201 — C Programming

A Rule-Driven Approach to Streamline Task Flow, Eliminate Delays, and Improve Team Productivity

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Introduction

- Enterprise task management coordinates work across teams, tools, and processes.
- Manual routing and unclear ownership frequently cause delays and rework.
- Bottlenecks form when tasks queue at overloaded users, approvers, or stages.
- A rule-based engine can automatically route, prioritize, and escalate tasks.
- Rules encode organizational policies in a transparent, editable form.
- Real-time dashboards surface where work is stuck and why.
- Goal: reduce cycle time, balance workloads, and improve throughput.



Problem Statement

- Task assignment in most enterprises is manual and reactive.
- Workload is unevenly distributed — a few users become bottlenecks.
- High-priority tasks are missed because rules exist only in people's heads.
- Escalation and approval delays extend overall cycle time.
- Managers lack visibility into where and why work is stuck.
- Result: missed SLAs, higher costs, and reduced customer satisfaction.
- Need: an automated, rule-driven workflow with real-time monitoring.



Objectives

- Design a rule-based workflow engine for enterprise task management.
- Model tasks, users, roles, priorities, and SLA deadlines.
- Define configurable rules for routing, prioritization, and escalation.
- Automatically detect and flag bottleneck stages and overloaded users.
- Build a dashboard to visualize task flow, workload, and delays.
- Compare manual vs rule-based routing on cycle time and throughput.
- Demonstrate measurable reduction in bottlenecks and waiting time.



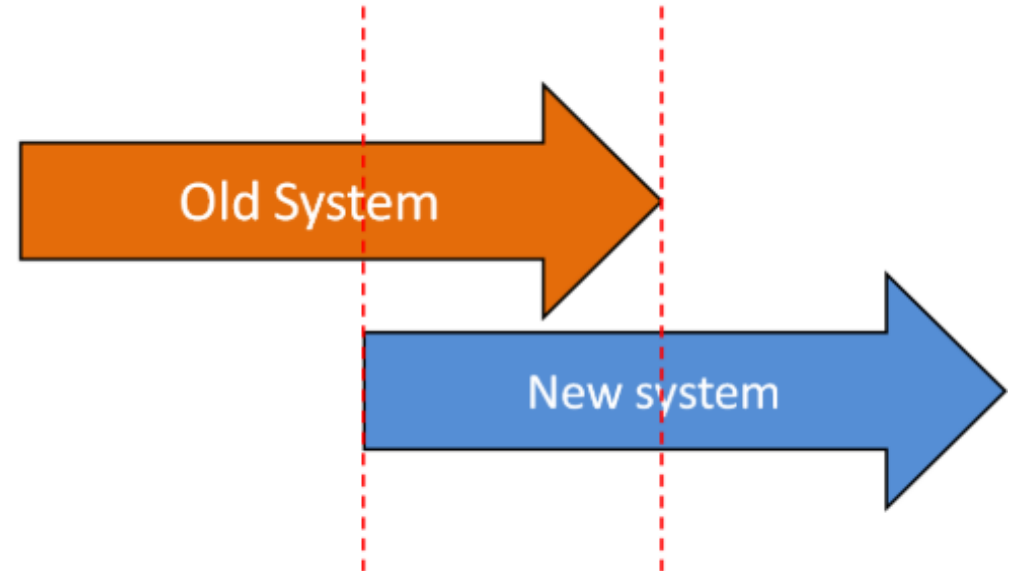
Existing System

- Tasks are assigned manually.
- Workload is not shared equally.
- Important tasks may be delayed.
- No automatic alerts for deadlines.
- Productivity is reduced due to bottlenecks.



Proposed System

- Tasks are assigned automatically.
- Urgent tasks get higher priority.
- Workload is balanced among employees.
- Delays are detected early.
- Improves productivity and saves time.



Module 1 — Workflow & Task Management

- Create, update, and close tasks with title, description, and attachments.
- Assign attributes: priority, category, due date, required role, and SLA.
- Model workflow stages: New → In Progress → Review → Done.
- Maintain user profiles with roles, skills, and current workload.
- Track task history — who did what, when, and how long each stage took.
- Expose APIs for creating tasks, updating status, and querying queues.
- Foundation layer that feeds data into the rule engine and dashboard.

Module 2 — Rule-Based Optimization & Dashboard

- Uses IF–THEN rules.
- Assigns tasks to the right employee.
- Prioritizes urgent tasks.
- Detects bottlenecks automatically.
- Dashboard shows task progress and workload.

Future Scope

- Add AI for smart task assignment.
- Predict bottlenecks in advance.
- Support multiple departments.
- Develop a mobile app.
- Integrate with email and ERP systems.



Literature Review

Authors & Year	Title	Key Contribution	Relevance to Project	Authors & Year
Yu & Yao (2026)	<i>Agents and Workflows</i>	Explores intelligent agents and workflow management using large language models for task execution and automation.	Provides the conceptual foundation for agent-based workflow optimization and intelligent task management.	Yu & Yao (2026)
Shoba et al. (2026)	<i>Hierarchical ML Ensemble Engine for Dynamic Project Management Optimization and Resource Intelligence</i>	Proposes an ML ensemble framework for optimizing project management and resource allocation dynamically.	Demonstrates techniques for improving resource utilization and workflow efficiency in enterprise environments.	Shoba et al. (2026)
Kosińska et al. (2026)	<i>Rule-based Execution of Management Actions in Cloud-native Environments</i>	Introduces a rule-based approach for automating management actions in cloud-native systems.	Supports the use of constraint-based and rule-driven mechanisms for workflow execution and bottleneck reduction.	Kosińska et al. (2026)
Yandamuri (2026)	<i>AI-Enabled Workflow Automation and Predictive Analytics for Enterprise Operations Management</i>	Presents AI-driven workflow automation integrated with predictive analytics for enterprise operations.	Highlights the role of AI in predicting performance trends and enhancing operational decision-making.	Yandamuri (2026)

Thank You

Questions & Discussion